



Determinants and Prediction of Global Economic Growth: An Ensemble Model Approach

being a dissertation submitted in fulfilment of the
requirements for the degree of

Master of

Artificial Intelligence and Data Science

in the University of Hull

by

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December 2025

Abstract

Forecasting economic growth with high precision remains a fundamental challenge for governments, central banks, and the private sector, given the non-linear and dynamic nature of macroeconomic indicators. This dissertation addresses this complexity by developing a robust data science framework capable of navigating the complex dynamics inherent in macroeconomic data. Using a balanced panel of 60 countries stratified by income level from the World Bank's World Development Indicators (2000–2021), this study implements a rigorous pipeline integrating hybrid imputation and Recursive Feature Elimination (RFE) to isolate 20 structural predictors. Five algorithms were evaluated: Ridge Regression, Support Vector Regression (SVR), Random Forest (RF), Long Short-Term Memory (LSTM), and a Hybrid Ensemble (LSTM+RF). The results demonstrate that the novel Hybrid Ensemble achieved superior predictive accuracy with a Root Mean Squared Error (RMSE) of 0.0452, outperforming LSTM (0.0453), RF (0.0458), SVR (0.0460), and the linear baseline Ridge (0.0464). Furthermore, the application of SHapley Additive exPlanations (SHAP) revealed critical structural heterogeneity, identifying that drivers such as fertility and infrastructure access exhibit polarity reversals depending on a nation's development stage. These findings validate the efficacy of hybrid machine learning architectures for economic forecasting and provide a transparent framework for decoding the context-dependent determinants of growth, advocating for the design of stage-specific economic policies rather than universal approaches, whilst underscoring the need for context-aware artificial intelligence (AI) tools in policy and decision-making.

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Chapter 1 Introduction and Background

Predicting economic growth is a fundamental challenge in macroeconomics, traditionally approached through linear econometric models. Enhancing the accuracy of Gross Domestic Product (GDP) forecasting is widely recognised as imperative not only for government budget planning and central bank counter-cyclical policies but also for the private sector to navigate market volatility (Elliott & Timmermann, 2016). However, the global economy acts as a complex system characterised by non-linear interactions and structural volatility (Wooldridge, 2020). The proliferation of high-dimensional datasets offers an opportunity to apply machine learning (ML) techniques to capture these complexities, although it faces a dual challenge: the small dataset problem inherent to annual national statistics and the 'black box' nature of deep learning (DL) algorithms, which limits their interpretation for decision-makers (Arrieta et al., 2020). This trade-off between accuracy and interpretability creates a critical gap in the application of AI for economic strategy.

Growth theory has evolved from Solow's (1956) physical capital and Romer's (1986) endogenous models to multidimensional frameworks. Recent studies (Anesti et al., 2021; Liu et al., 2023; Vrontos et al., 2024) demonstrate that integrating institutional, social, environmental, and technological indicators enhances non-linear forecast accuracy, validating the utility of high-dimensional feature spaces for predicting global growth.

Standard forecasting methods face limitations in reconciling the need for temporal memory to capture historical trends and the flexibility to model non-linear structural interactions (Liu et al., 2025). Linear models exhibit high bias in complex environments (Hoerl & Kennard, 1970). Conversely, DL architectures require large volumes of data to generalise effectively (Hochreiter & Schmidhuber, 1997), making them prone to overfitting in annual macroeconomic datasets with limited samples. Furthermore, exogenous shocks, such as the COVID-19 pandemic, introduce concept drift that renders static models obsolete by altering the underlying data distribution (Vrontos et al., 2024). Consequently, there is a need for unified frameworks that effectively integrate temporal dynamics with structural analysis while maintaining interpretability (Liu et al., 2025).

These constraints have prompted a comparative evaluation of ML architectures for GDP prediction. In high-dimensional environments, Medeiros et al. (2021) demonstrated that Ridge Regression consistently outperformed other linear regressions and factor models by effectively handling multicollinearity, establishing it as a robust baseline. To address non-linear dynamics, kernel-based and ensemble methods have been widely adopted. Anesti et al. (2021) and Richardson et al. (2020) validated the efficacy of SVR in forecasting GDP growth, attributing its success to its robustness in handling high-dimensional survey data. Furthermore, Medeiros et al. (2021) and Coulombe et al. (2020) identified RF as a top-performing algorithm for macroeconomic forecasting, attributing its success to the variance reduction. Regarding DL, Chen et al. (2025) demonstrated that LSTM outperformed advanced architectures such as Recurrent Neural Networks and Gated Recurrent Unit by capturing long-term temporal dependencies. Additionally, current literature

(Liu et al., 2025; Perone & Zambrano-Monserrate, 2025) validates that hybrid frameworks consistently outperform isolated algorithms.

Crucially, while the predictive superiority of ML and hybrid architectures is well-established, existing literature exhibits distinct gaps addressed by this study. First, research has predominantly focused on single advanced economies (Anesti et al., 2021; Richardson et al., 2020) or specific regional clusters (Perone & Zambrano-Monserrate, 2025), thereby neglecting the structural heterogeneity inherent in a global dataset stratified by development stages. Second, while advanced DL approaches achieve high accuracy (Chen et al., 2025; Liu et al., 2023), they often function as 'black boxes,' prioritising aggregate error reduction over the interpretability required to identify how growth determinants might exhibit polarity reversals between income levels. Consequently, there is a scarcity of research that effectively combines hybrid architectures with Explainable AI (XAI) to decode these state-dependent dynamics globally, leaving a gap in the provision of context-aware tools for differentiated policymaking.

Given this context, this dissertation aims to develop a data science pipeline to predict global economic growth and decode its determinants. The specific objectives are: 1) to evaluate and compare the predictive efficacy of five distinct architectures: Linear (Ridge), Geometric (SVR), Ensemble (RF), Deep Learning (LSTM), and Hybrid Ensemble (LSTM+RF), and 2) to implement XAI via SHAP values to uncover structural heterogeneity across income levels.

This study contributes to the field of Applied Data Science by validating a Hybrid Ensemble architecture (LSTM+RF). It demonstrates that combining the low temporal variance of recurrent networks with the low structural bias of decision trees yields superior generalisation during volatile periods (2018–2021). Additionally, SHAP values provide empirical evidence that the drivers of growth are state-dependent, challenging universalist economic models.

Chapter 2 Methodology

2.1 Data Source and Initial Selection

2.1.1 Data Source

The dataset employed was derived from the World Bank's World Development Indicators (WDI), recognised as the premier repository for cross-country comparable statistics on development, encompassing over 1,500 time-series indicators and covering 217 economies (World Bank, 2025). The World Bank organises these indicators into six thematic pillars: Poverty and Inequality, People, Environment, Economy, States and Markets, and Global Links; and stratifies economies into four distinct income levels (high, upper-middle, lower-middle, and low income), a classification utilised in this study to uncover structural heterogeneity in growth determinants.

2.1.2 Variable Preselection

To capture the multidimensional nature of growth, an initial set of 60 variables was curated, selecting 10 indicators for each World Bank thematic pillar based on economic theories such as the Neoclassical Model (Solow, 1956), the Endogenous Growth Model (Romer, 1986), and the Expenditure Approach (Keynes, 1936); whilst social and environmental metrics were aligned with the United Nations (2015) Sustainable Development Goals framework. The initial variable selection is detailed in Appendix 1.

2.1.3 Temporal Scope

The analysis period was defined as 2000–2021. As illustrated in Figure 1, a historical data density analysis revealed that availability peaks from 2000 onwards. The cut-off at 2021 was necessitated by the reporting lag inherent in indicators.

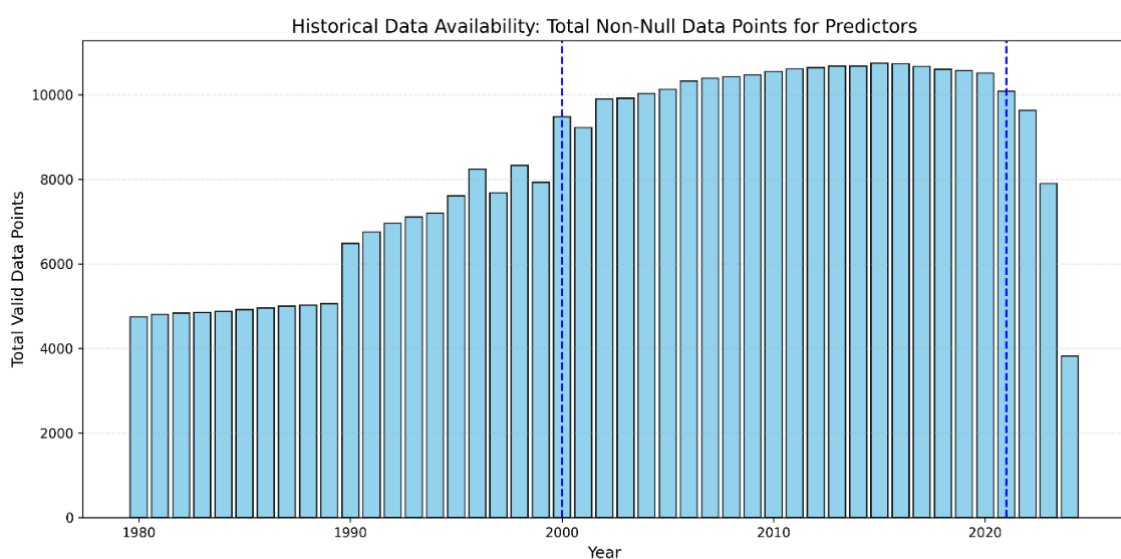


Figure 1: Annual count of non-null predictor values (1980–2024).

2.1.4 Country Selection

To ensure a robust and representative sample, a two-stage selection algorithm was implemented to select countries. First, a data quality filter: countries with less than 75% data availability for the selected variables were excluded to minimise the bias introduced by excessive imputation. Second, an economic relevance filter: from the remaining candidates, the top 15 economies (by Total GDP in 2021) were selected within each income level. This process resulted in a balanced panel of 60 countries (shown in Table 1), ensuring global representativeness.

Table 1: Selected countries

	HIGH INCOME	UPPER MIDDLE INCOME	LOWER MIDDLE INCOME	LOW INCOME
1	United States	China	India	Sudan
2	Japan	Brazil	Nigeria	Congo, Dem. Rep.
3	Germany	Mexico	Egypt, Arab Rep.	Uganda
4	United Kingdom	Türkiye	Pakistan	Mali
5	France	Indonesia	Philippines	Mozambique
6	Italy	Argentina	Viet Nam	Burkina Faso
7	Canada	Iran, Islamic Rep.	Bangladesh	Syrian Arab Republic
8	Korea, Rep.	Thailand	Morocco	Chad
9	Australia	Malaysia	Uzbekistan	Niger
10	Russian Federation	South Africa	Sri Lanka	Madagascar
11	Spain	Colombia	Kenya	Rwanda
12	Netherlands	Peru	Angola	Malawi
13	Switzerland	Kazakhstan	Ghana	Sierra Leone
14	Saudi Arabia	Algeria	Tanzania	Togo
15	Poland	Iraq	Côte d'Ivoire	Burundi

2.2 Target Variable Construction

To model economic performance and ensure cross-country comparability, the target variable (Y) was defined as the annual growth rate of GDP per capita, based on Purchasing Power Parity (PPP) in constant 2017 international dollars (WDI Code: NY.GDP.PCAP.PP.KD). Following standard econometric practice (Wooldridge, 2020), this was calculated as equation (1) using the logarithmic difference:

$$Y_{it} = \ln(GDP_{it}) - \ln(GDP_{i,t-1}) \quad (1)$$

where i denotes the country and t represents the year. This transformation helps stabilise ML by converting the series into a stationary sequence and producing a symmetric distribution of the target variable.

2.3 Data Pre-processing

2.3.1 Data Imputation

Given the prevalence of missing values, a two-stage imputation strategy was implemented. First, Linear Interpolation was applied to fill gaps within existing time series for each country. Second, a K-Nearest Neighbours (KNN) algorithm was utilised to estimate the remaining missing values based on cross-country similarities. This strategy was validated through a masking experiment, achieving an R^2 score of 0.91, confirming the preservation of statistical integrity.

2.3.2 Temporal Alignment

In economic forecasting, applying a time lag is good practice to mitigate endogeneity and look-ahead bias (Vrontos et al., 2024). In this study, the predictor variables are shifted by one year. Consequently, the models utilise values from year $t-1$ to predict growth in year t .

2.3.3 Standardisation

The pre-selected indicators were normalised using Z-score standardization, scaling each variable in order to have a mean of 0 and a standard deviation of 1, thus ensuring a fair comparison between attributes during model training.

2.4 Feature Selection

2.4.1 Recursive Feature Elimination (RFE)

To reduce the dimensionality and multicollinearity represented by the initial set of 60 variables, RFE was employed, a wrapper method originally proposed by Guyon et al. (2002). Using a linear estimator as the base, the RFE algorithm iteratively eliminated the least significant features based on coefficient weights. Table 2 shows the 20 most relevant predictors retained, categorised according to the World Bank's thematic pillars. This selection keeps the multi-dimensional view of development.

Table 2: Indicators selected through RFE

Selected Predictors per Thematic Pillar	
Theme: Environment Access to electricity Access to clean fuels and technologies for cooking People using at least basic sanitation services People using at least basic drinking water services Energy use Carbon dioxide emissions PM2.5 air pollution, mean annual exposure	Theme: Economy Urban population growth Gross savings Agriculture, forestry, and fishing, value added Industry (including construction), value added
	Theme: Poverty and Inequality Poverty headcount ratio at national poverty lines
	Theme: States and Markets Domestic credit to private sector Government Effectiveness: Estimate
	Theme: Global Links Exports of goods and services Imports of goods and services
Theme: People Prevalence of undernourishment Life expectancy at birth Labor force participation rate Fertility rate	

2.5 Predictive Models

Five distinct architectures were selected to benchmark linear stability against non-linear complexity. Ridge serves as a robust baseline, contrasted by SVR and RF which capture non-linearities. LSTM addresses specific temporal dependencies, while the Hybrid model tests ensemble generalisation. Hyperparameter optimisation strategies were adapted to each model family: the classical algorithms used Grid Search with Time Series Cross-Validation (3 splits) to prevent data leakage, whilst the DL model applied a temporal hold-out method (20% validation). The dataset was partitioned chronologically: 2001–2017 for training and 2018–2021 for testing.

2.5.1 Ridge Regression

The baseline model for this study was Ridge Regression, a linear approach that applies L2 regularisation to reduce overfitting by penalising large coefficients (Hoerl & Kennard, 1970). While its reliance on linear assumptions restricts the capture of complex dynamics, it remains a robust standard for high-dimensional data. The parameter that controls the intensity of the regularisation (α) was optimised during tuning.

2.5.2 Support Vector Regression

SVR was implemented to capture non-linear relationships by constructing a regression hyperplane (Cortes & Vapnik, 1995). Although this method is computationally intensive and sensitive to noise, it offers high flexibility. To manage these trade-offs and prevent overfitting, the regularisation parameter (C) and the tolerance margin (ϵ) were tuned to balance model complexity with error tolerance.

2.5.3 Random Forest Regression

RF was selected for its ability to model complex interactions through an ensemble of independent trees (Breiman, 2001). A critical limitation is its inability to extrapolate values beyond the range of the training data; however, it excels at variance reduction. Overfitting was prevented by restricting tree depth (*max_depth*) and the minimum sample requirement per leaf node (*min_samples_leaf*), forcing the model to learn general patterns rather than memorising.

2.5.4 Long Short-Term Memory Network

Addressing the sequential nature of GDP growth, a LSTM network was designed to retain long-term information (Hochreiter & Schmidhuber, 1997). While this architecture introduces 'black box' opacity and higher data requirements, it is superior in handling temporal dependencies. To mitigate the risk of overfitting, the architecture comprised two LSTM layers (64, 32 neurons) incorporating Dropout (0.2) and Early Stopping.

2.5.5 Hybrid Model

Based on the hypothesis that combining diverse algorithms improves performance (Dietterich, 2000), a hybrid model merged LSTM predictions (temporal patterns) with RF (ensemble generalisation). Although this increases architectural complexity and computational cost, it effectively offsets the weaknesses of isolated

models. Consequently, this strategy is particularly well-suited for leveraging both sequential dynamics and cross-sectional interactions.

2.6 Model Evaluation

Model performance was evaluated on the test set using three metrics: RMSE as the primary metric, chosen for rigorously penalizing the significant deviations characteristic of economic crises; Mean Absolute Error (MAE) as a complementary and more interpretable measure of error; and the Coefficient of Determination (R^2) to assess the proportion of variance explained by the predictors. Additionally, the generalisation gap—defined as the percentage difference between the training RMSE and the test RMSE—was monitored to quantify overfitting.

2.7 Model Interpretation

To decode the specific determinants of GDP growth, SHAP values were calculated specifically for the model identified as the top performer. This approach, based on cooperative game theory, assigns a contribution value to each variable, allowing the identification of the variables with the greatest impact on the model (Lundberg & Lee, 2017).

Chapter 3 Results

3.1 Model Performance Comparison

The predictive accuracy of the six algorithms was evaluated on the held-out test set (2018–2021) using the RMSE as the primary selection metric. Table 3 presents the performance leader board.

Table 3: Evaluation metrics of the models employed.

Model Comparison (Performance on Test Set):			
Model	RMSE	MAE	R2
Hybrid	0.0452	0.0320	-0.0237
LSTM	0.0453	0.0326	-0.0287
RF	0.0458	0.0322	-0.0544
SVR	0.0460	0.0320	-0.0774
Ridge	0.0464	0.0328	-0.0821

The proposed Hybrid Ensemble (LSTM+RF) recorded the lowest error (RMSE: 0.0452), followed closely by the LSTM network. The independent RF occupied the middle tier of performance. Conversely, the linear baseline (Ridge) produced the highest error (RMSE: 0.0464), lagging behind the non-linear and hybrid approaches in predictive accuracy.

3.2 Generalisation and Overfitting Analysis

The generalisation capacity of each model was assessed by comparing RMSE values across training and test sets, with the percentage gap serving as an indicator of potential overfitting. A smaller gap indicates better stability to unseen data. As shown in Table 4, the LSTM network demonstrated the smallest divergence (12.2%), followed by the Hybrid Ensemble (17.0%) and Ridge (18.4%). In contrast, RF and SVR models displayed the largest gaps (22.0% and 24.0% respectively). Ultimately, the Hybrid model offers the optimal trade-off. Although it showed a moderate gap (17.0%)—indicating a high capacity to learn from training data—this translated directly into predictive power, resulting in the best RMSE (0.0452) on unseen data.

Table 4: Overfitting analysis based on RMSE gap between training and test sets.

Overfitting Analysis			
Model	Train_RMSE	Test_RMSE	Gap (%)
LSTM	0.0403	0.0453	12.2
Hybrid	0.0386	0.0452	17.0
Ridge	0.0392	0.0464	18.4
RF	0.0376	0.0458	22.0
SVR	0.0374	0.0463	24.0

3.3 Predictive Behaviour and Variance

Based on the balance between accuracy and stability, the Hybrid Ensemble is identified as the overall best-performing architecture. However, despite this superiority in relative metrics, it yielded negative R^2 value (-0.024). As illustrated in Figure 2, there is a marked divergence between the predicted trajectory and the actual observations.

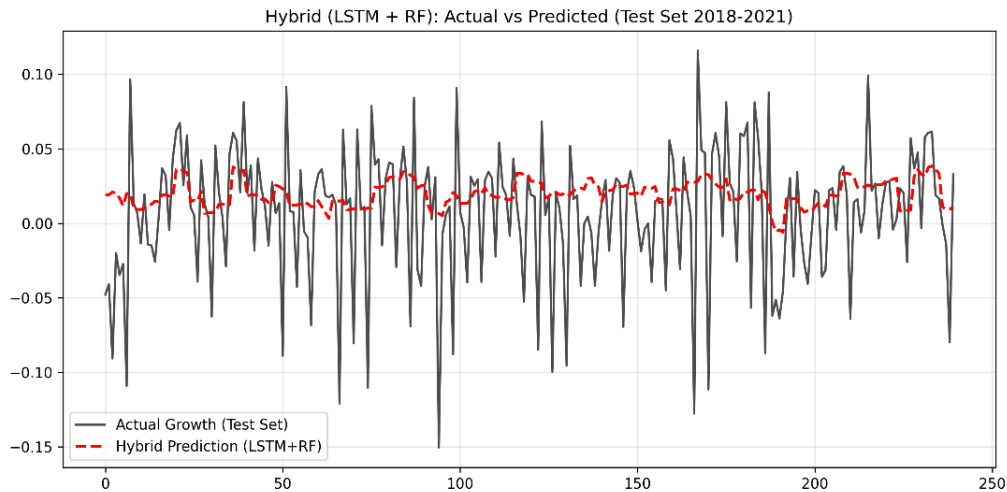


Figure 2: Hybrid model predictions vs. actual growth (test set).

Hybrid model generates a conservative, low-variance prediction (red line), concentrated around the central trend of historical growth. In contrast, the actual data (black line) exhibits volatility. While the model accurately predicts the behaviour based on fundamental indicators and successfully minimises the RMSE, the negative R^2 value reflects its inability to capture the high-variance present during the 2018–2021 period.

3.4 Determinants of Growth

The SHAP-based analysis of feature contributions reveals notable variations in the drivers of predicted growth across income groups. Figure 3 presents a comparative heatmap where colour intensity reflects the mean impact on growth. SHAP values appear numerically small since they represent marginal additive components of the GDP growth rate—a metric which itself typically ranges within a narrow decimal band; thus, even minor decimal deviations signify substantial economic impacts. Positive values imply the feature acts as a 'booster' (pushing the predicted growth rate higher), whereas a negative value indicates the feature acts as a 'drag' (exerting a downward pressure on the growth rate).

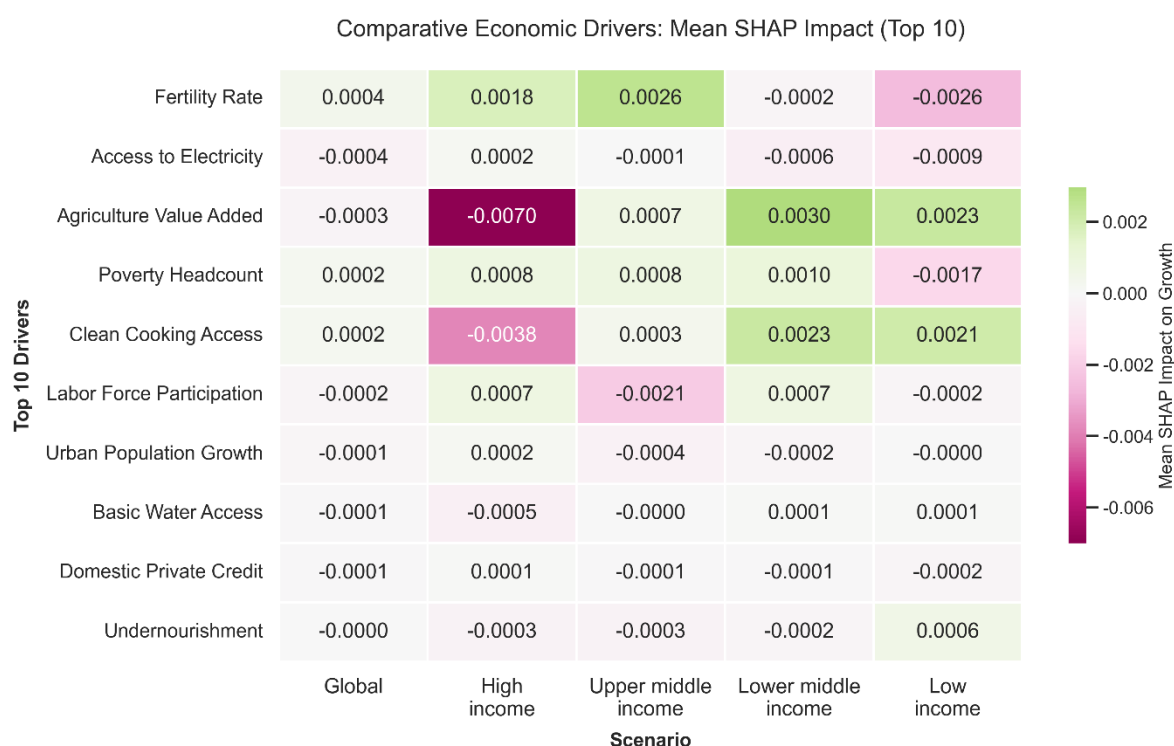


Figure 3: Heatmap of the top 10 drivers across income groups (SHAP Analysis).

As illustrated in Figure 3, the four most significant determinants in the global scenario—Fertility Rate, Access to Electricity, Agricultural Value Added, and Poverty Headcount—exhibit remarkably low mean SHAP values ($< |0.0004|$), appearing as neutral drivers. When analysed by income level, distinct patterns emerge. Fertility Rate, Poverty Headcount, and Access to Electricity exert a positive impact for more developed countries but act as a drag on poorer economies. Conversely, Agricultural Value and Access to Clean Fuels serve as positive drivers, except for high-income countries, where the values highlight the strongest negative magnitudes observed in the dataset (-0.0070 and -0.0038, respectively). Finally, Labor Force Participation displays a localised negative effect (-0.0021) only within the upper-middle income group. These results highlight the heterogeneity of growth drivers across different strata, reinforcing the need for context-sensitive modelling approaches.

Chapter 4 Discussion

4.1 Evaluation of Models

The performance hierarchy observed provides deep insight into the underlying structure of economic growth data. The inferiority of Ridge confirms that economic growth is fundamentally a nonlinear process, necessitating architectures capable of mapping complex interactions, such as SVR and RF. However, the superior performance of the Hybrid Ensemble (LSTM + RF) over independent models validates the theoretical premise of Stacking Generalization (Wolpert, 1992). By combining the sequence recognition of LSTM with the variance reduction of RF, the hybrid framework effectively mitigates the specific weaknesses of each algorithm—noise memorisation in RF and high data dependency in LSTM. This aligns with recent findings by Liu et al. (2025), confirming that hybrid DL architectures excel in time series prediction by leveraging complementary learning strategies.

4.2 Interpreting Negative R^2

The negative R^2 values observed across all models provide quantitative evidence of Dataset Shift (Quiñonero-Candela et al., 2009). As illustrated in Figure 4, the target distribution during the COVID-19 shock differed fundamentally from the training period (2001–2017). The training distribution (blue) approximates a normal curve centred on positive growth, whereas the shock distribution (red) exhibits a significant negative mean shift.

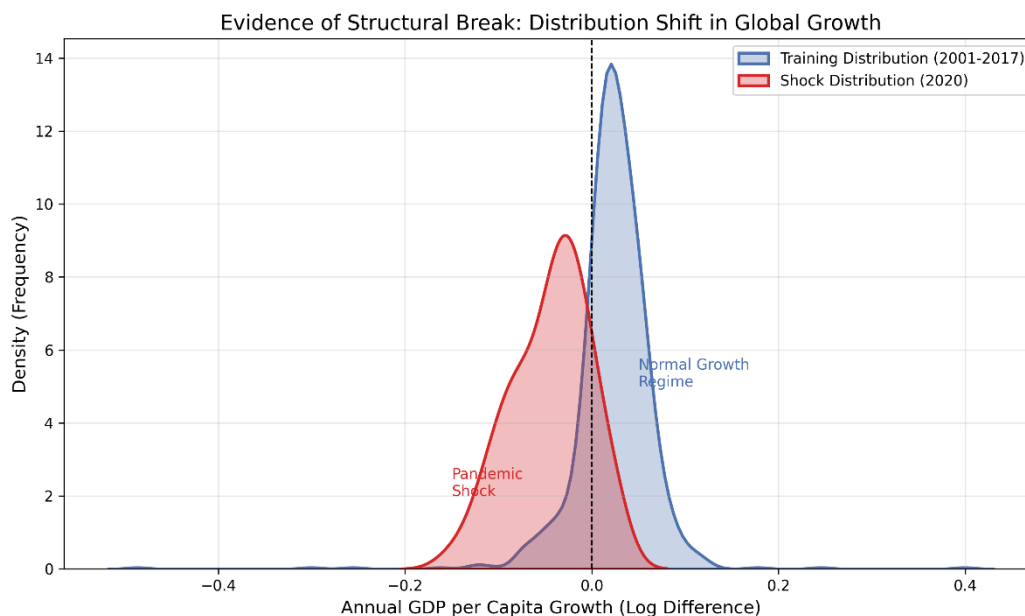


Figure 4: Evidence of distribution shift (training data vs. pandemic shock).

This aligns with Schölkopf et al. (2021), who establish that statistical ML models struggle to generalise when the underlying environmental rules change. Consequently, the architectures failed to extrapolate the

magnitude of the recession beyond the stable training range. However, the Hybrid model's minimised RMSE suggests it successfully modelled the structural growth potential, even though the negative R^2 reflects the huge variance of the exogenous shock.

4.3 Interpretability and Structural Heterogeneity

The application of SHAP provided a layer of interpretability essential for validating the trustworthiness of the 'black box' models (Arrieta et al., 2020). SHAP values reveal decisive structural patterns that validate the hypothesis of heterogeneity.

According to Bloom et al. (2003), fertility in developed countries can mitigate population ageing and renew the labour force; whereas in low-income backgrounds, higher fertility results in a negative contribution as it increases the dependency burden and limits savings and investment in human capital, thus hindering growth. In low-income countries, high levels of poverty reflect structural constraints that reduce productivity and hinder growth, while at other income levels, relative increases in poverty often accompany early recovery phases and are therefore correlate positively with growth (Cerra et al., 2021). In high-income economies, electricity access principally enhances efficiency and productivity. Conversely, in other contexts, expansion frequently outpaces installed capacity, coinciding with infrastructural bottlenecks, grid unreliability and high costs, generating short-term constraints manifested as negative contributions to economic growth (IEA, 2025).

Aligned with the theory of Structural Transformation (Herrendorf et al., 2013), for low- and middle-income nations, agricultural productivity is the primary engine of surplus, underpinning employment, food security, and exports. However, for high-income nations, a reliance on agriculture signals a failure to transition towards high-productivity sectors (e.g., finance and business services), acting as a drag on growth. Access to clean fuels and technologies for cooking represents a massive productivity leap in developing nations, releasing labour time and improving health. In rich nations, this variable is saturated, so any variance likely correlates with non-industrial rurality, hence the negative signal (IEA, 2025).

Finally, labour force participation emerges as a significant negative driver uniquely for the upper-middle income group. This provides empirical evidence for the Middle-Income Trap (Gill & Kharas, 2007). Economies in this transitional stage can no longer sustain growth merely by adding more workforce; they face diminishing returns on labour if they fail to shift towards innovation-led growth, a friction that neither low nor high-income groups experience to the same degree.

4.4 Limitations

Despite the robustness of the hybrid approach, this study faces inherent constraints. Firstly, the dataset remains small for the LSTM model, restricting its capacity to learn deeper temporal abstractions compared to large-scale datasets, a challenge intrinsic to neural networks (Hochreiter & Schmidhuber, 1997) and frequently encountered in macroeconomic deep learning applications (Chen et al., 2025). Secondly, while the KNN imputation ($R^2=0.91$) was statistically valid, it introduces synthetic noise that may smooth out volatility in data-poor regions. This potential to underestimate risks aligns with the classical problem of measurement error bias described by Wooldridge (2020). Finally, the predictive failure during the 2020 shock highlights the limitation of supervised models to anticipate exogenous structural breaks that lie outside the learned distribution, a fundamental issue of dataset shift (Quiñonero-Candela et al., 2009) consistent with the specific challenges of forecasting during pandemics noted by Vrontos et al. (2024).

4.5 Future Research

Future work leveraging higher-frequency data (e.g., quarterly) should explore Transformer architectures, specifically Temporal Fusion Transformers (TFT). Recent studies by Lim et al. (2021) suggest that TFTs offer superior interpretability and performance in multi-horizon forecasting by employing attention mechanisms to weigh the importance of different time steps dynamically. Additionally, integrating unstructured data (e.g., financial news or satellite imagery) could provide real-time signals to correct predictions during structural breaks like pandemics. Text-based indicators can capture real-time economic sentiment shifts that precede official statistical releases through Natural Language Processing (NLP) (Barbaglia et al., 2023). Meanwhile, including satellite luminosity data can improve GDP nowcasting accuracy, especially in developing economies by correlating night-time lights with economic activity (Beyer et al., 2021).

Chapter 5 Conclusion

This research addressed the complexity of forecasting global economic growth by evaluating five ML architectures. The empirical results established that traditional linear methods (Ridge Regression) are insufficient for capturing non-linear growth dynamics. The central finding is the superiority of the proposed Hybrid Ensemble (LSTM+RF), which achieved the lowest error (RMSE: 0.0452) by integrating temporal memory with structural decision trees. Furthermore, the analysis of negative R^2 values quantified the impact of the COVID-19 structural break.

These findings offer distinct implications for stakeholders. For global policymakers and development agencies, the empirical evidence of structural heterogeneity challenges universalist prescriptions, advocating instead for context-aware strategies that adapt to a nation's development stage. Simultaneously, for economic strategists in both public and private sectors, the contrast between the model's precision during stability and its breakdown during the 2020 shock delivers a cautionary mandate: while algorithmic forecasting is a powerful tool for routine planning, it requires qualitative human oversight during unprecedented structural breaks.

Three contributions to applied data science are identified. First, this study establishes a rigorous pipeline integrating RFE and temporal validation for robust macroeconomic modelling. Second, it validates a Hybrid Ensemble architecture, demonstrating that combining the low bias of trees with the memory of recurrent networks mitigates overfitting more effectively than individual algorithms. Finally, it demonstrates the utility of SHAP for decoding the 'black box' of growth, bridging the gap between predictive accuracy and the interpretability required for economic strategy.

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Appendix

Appendix 1: Initial selection of 60 variables

Theme	Variable Name
Poverty and Inequality	Unemployment, youth total (% of total labor force ages 15-24) Access to electricity (% of population) Access to clean fuels and technologies for cooking (% of population) Prevalence of undernourishment (% of population) Rural population (% of total population) Current health expenditure (% of GDP) People using at least basic sanitation services (% of population) People using at least basic drinking water services (% of population) Mortality rate, under-5 (per 1,000 live births) Poverty headcount ratio at national poverty lines (% of population)
People	Population growth (annual %) Life expectancy at birth, total (years) School enrollment, tertiary (% gross) Government expenditure on education, total (% of GDP) Labor force participation rate, total (% of total population ages 15+) Age dependency ratio (% of working-age population) Urban population growth (annual %) Fertility rate, total (births per woman) School enrollment, primary (% gross) Adolescent fertility rate (births per 1,000 women ages 15-19)
Environment	Energy use (kg of oil equivalent per capita) Carbon dioxide (CO2) emissions excluding LULUCF per capita Renewable energy consumption (% of total final energy consumption) Arable land (% of land area) Forest area (% of land area) Agricultural land (% of land area) Agricultural irrigated land (% of total agricultural land) Total natural resources rents (% of GDP) PM2.5 air pollution, mean annual exposure Annual freshwater withdrawals, total (% of internal resources)
Economy	Inflation, consumer prices (annual %) Real interest rate (%) Gross capital formation (% of GDP) Gross savings (% of GDP) General government final consumption expenditure (% of GDP) Unemployment, total (% of total labor force) Broad money (% of GDP) Domestic credit to private sector (% of GDP) Agriculture, forestry, and fishing, value added (% of GDP) Industry (including construction), value added (% of GDP)
States and Markets	New business density (new registrations per 1,000 people) Tax revenue (% of GDP) Military expenditure (% of GDP) Interest rate spread (lending rate minus deposit rate, %) Mobile cellular subscriptions (per 100 people) Individuals using the Internet (% of population) Control of Corruption: Estimate Political Stability and Absence of Violence/Terrorism: Estimate Government Effectiveness: Estimate Market capitalization of listed domestic companies (% of GDP)
Global Links	Trade (% of GDP) Exports of goods and services (% of GDP) Imports of goods and services (% of GDP) Foreign direct investment, net inflows (% of GDP) Personal remittances, received (% of GDP) Net official development assistance received (% of GNI) External debt stocks (% of GNI) Tariff rate, applied, simple mean, all products (%) International tourism, number of arrivals Merchandise trade (% of GDP)